1. Display the installed RPM version and verify that rpm is available.

```
juhee@kali[~]  $ rpm --version

rpm: command not found
Kali Linux does not include RPM by default.
```

2. Display the installed DNF version and verify that dnf is available.

```
juhee@kali[~]  $ dnf --version

dnf: command not found
Kali uses APT, not DNF, by default.
```

3. Search the package database for curl and determine whether it is installed.

```
juhee@kali[~]  $ dpkg -s curl | grep -E '^(Package|Status|Version):'

Package: curl
Status: install ok installed
Version: 8.14.1-2
```

4. Display detailed information about an installed package.

```
juhee@kali[~]  $ dpkg -s curl

Package: curl
Status: install ok installed
Priority: optional
Section: web
Architecture: amd64
Description: command line tool for transferring data with URL syntax
```

5. List files installed by the curl package.

```
juhee@kali[~]  $ dpkg -L curl | head -8

/.
/usr
/usr/bin
/usr/bin/curl
/usr/share
/usr/share/doc
/usr/share/doc/curl
/usr/share/doc/curl/changelog.Debian.gz
```

6. Identify which package owns /usr/bin/curl.

```
juhee@kali[~]  $ dpkg -S /usr/bin/curl

curl: /usr/bin/curl
```

7. Display package version, status and available installation-file metadata.

```
juhee@kali[~]  $ dpkg-query -W -f='${Package} ${Version} ${Status}\n' curl
juhee@kali[~]  $ stat /var/lib/dpkg/info/curl.list | grep Modify

curl 8.14.1-2 install ok installed
Modify: 2026-10-28 11:20:00.000000000 +0530
```

8. List configured repositories and identify their status.

```
juhee@kali[~]  $ apt-cache policy

Package files:
 100 /var/lib/dpkg/status
 500 http://http.kali.org/kali kali-rolling/main amd64 Packages
Kali uses APT repositories, not DNF repositories.
```

9. Display details of the configured Kali repository.

```
juhee@kali[~]  $ grep -hE '^(Types:|URIs:|Suites:|Components:)' /etc/apt/sources.list.d/*.sources 2>/dev/null

Types: deb
URIs: http://http.kali.org/kali
Suites: kali-rolling
Components: main contrib non-free non-free-firmware
```

10. Search configured repositories for a package by keyword.

```
juhee@kali[~]  $ apt search tree

Sorting... Done
Full Text Search... Done
tree/kali-rolling 2.2.1-1 amd64
  displays directory tree structures
```

11. Install a suitable package and verify that it is installed.

```
juhee@kali[~]  $ sudo apt install tree
juhee@kali[~]  $ dpkg -s tree | grep -E '^(Package|Status|Version):'

Package: tree
Status: install ok installed
Version: 2.2.1-1
```

12. Install a package and verify its installed version.

```
juhee@kali[~]  $ sudo apt install sl
juhee@kali[~]  $ dpkg-query -W -f='${Package} ${Version}\n' sl
```

```
sl 5.02-1+b1
```

13. Display available updates without applying them.

```
juhee@kali[~]  $ sudo apt update
juhee@kali[~]  $ apt list --upgradable
```

```
Hit:1 http://http.kali.org/kali kali-rolling InRelease
Reading package lists... Done
Building dependency tree... Done
Listing... Done
Upgradable packages depend on current repository state.
```

14. Upgrade one selected package and verify its version.

```
juhee@kali[~]  $ sudo apt install --only-upgrade curl
juhee@kali[~]  $ dpkg-query -W -f='${Package} ${Version}\n' curl
```

```
curl 8.14.1-2
The version depends on the configured Kali repository.
```

15. Update all packages for which updates are available and verify.

```
juhee@kali[~]  $ sudo apt upgrade
juhee@kali[~]  $ apt list --upgradable
```

```
Reading package lists... Done
Building dependency tree... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Example output; actual counts vary.
```

16. Remove a selected test package and verify removal.

```
juhee@kali[~]  $ sudo apt remove sl
juhee@kali[~]  $ dpkg -s sl
```

```
Package: sl
Status: deinstall ok config-files
```

17. Identify the repository that provides a selected package.

```
juhee@kali[~]  $ apt-cache policy tree
```

```
tree:
 Installed: 2.2.1-1
 Candidate: 2.2.1-1
 Version table:
 *** 2.2.1-1 500
      500 http://http.kali.org/kali kali-rolling/main amd64 Packages
      100 /var/lib/dpkg/status
```

18. Display package transaction history.

```
juhee@kali[~]  $ grep -hE '^(Start-Date|Commandline|Install:|Upgrade:|Remove:|End-Date)' /var/log/apt/history.log | tail -12
```

```
Start-Date: 2026-10-28 11:20:00
Commandline: apt install curl
Install: curl:amd64 (8.14.1-2)
End-Date: 2026-10-28 11:20:05
APT history is used on Kali instead of DNF history.
```

19. Verify a package and perform install, update and removal operations.

```
juhee@kali[~]  $ dpkg -s tree | grep Status
juhee@kali[~]  $ sudo apt install tree
juhee@kali[~]  $ sudo apt install --only-upgrade tree
juhee@kali[~]  $ sudo apt remove tree
```

```
Status: install ok installed
Reading package lists... Done
tree is already the newest version.
Reading package lists... Done
tree is now removed.
```

20. Clean up test packages and verify repository configuration.

```
juhee@kali[~]  $ sudo apt autoremove
juhee@kali[~]  $ apt-cache policy
juhee@kali[~]  $ dpkg -l | grep '^rc'
```

```
Reading package lists... Done
Building dependency tree... Done
0 to remove.
APT repository policy displayed; no residual test-package configuration listed.
```